

Architect's Brief

Online Marketplace Order Platform

1. Architecture Purpose

This brief prepares the Online Marketplace Order Platform for system architecture modeling. It identifies the likely system boundary, external participants, internal responsibilities, integrations, data ownership concerns, workflows, and quality drivers.

This brief is not a product requirements document and does not define implementation design or code structure.

2. System Boundary

The subject system is the **Online Marketplace Order Platform**.

The system owns checkout coordination, marketplace order creation, seller order splitting, payment coordination, fulfillment status tracking, cancellation handling, refund coordination, shipment status tracking, customer support order visibility, notification coordination, audit history, and integration status tracking.

The system does not own product catalog authoring, physical fulfillment operations, seller warehouse operations, payment network infrastructure, tax authority rules, carrier logistics, accounting, seller payout reconciliation, or marketing campaign management.

3. External Participants

Participant	Role in the Architecture
Buyer	Initiates checkout, views order status, requests cancellation or refund, and receives notifications.
Seller	Receives seller fulfillment orders, accepts or rejects fulfillment, provides shipment updates, and reports exceptions.
Customer Support Agent	Searches orders, reviews lifecycle history, and performs authorized support actions.
Marketplace Administrator	Configures marketplace rules, seller status, order policies, and operational controls.
Operations Manager	Monitors order flow, exceptions, seller response, and unresolved integration failures.
Payment Gateway	Provides payment authorization, capture, void, and refund operations.

Participant	Role in the Architecture
Tax Calculation Service	Provides transaction tax calculation.
Shipping Carrier Service	Provides shipping rates, labels, tracking events, and delivery status where configured.
Notification Provider	Sends buyer, seller, support, and operational notifications.
Seller System	Provides inventory confirmation, reservation, order import, or fulfillment update integration where configured.
Fraud Screening Service	Provides risk scoring or fraud review where configured.
Identity Provider	Authenticates sellers, support agents, administrators, or internal users where configured.

4. Candidate Internal System Responsibilities

Responsibility Area	Description
Checkout Coordination	Validates cart, shipping details, availability, totals, tax, fraud status, and payment authorization before order acceptance.
Order Orchestration	Creates marketplace orders, manages lifecycle transitions, and coordinates downstream order effects.
Seller Order Management	Splits marketplace orders into seller-specific fulfillment orders and tracks seller-level state.
Payment Coordination	Encapsulates payment authorization, capture, void, refund, and payment status handling.
Tax Coordination	Requests tax calculations and preserves tax results used at checkout.
Availability Coordination	Confirms item availability or reservation through local snapshots or seller integrations.
Fulfillment Coordination	Tracks seller acceptance, rejection, fulfillment updates, exceptions, and shipment readiness.
Shipment Tracking	Stores shipment data, imports carrier status, and exposes buyer/support

Responsibility Area	Description
	shipment visibility.
Cancellation Management	Validates and executes full or partial cancellations according to policy and fulfillment state.
Refund Management	Calculates, submits, records, and tracks full or partial refunds.
Notification Management	Sends order lifecycle notifications to buyers, sellers, support, and operations users.
Support Operations	Provides order search, lifecycle history, support notes, and approved support actions.
Policy Evaluation	Applies marketplace rules for payment timing, cancellation, refund, seller SLA, and support authority.
Audit Logging	Records checkout, order, payment, fulfillment, shipment, cancellation, refund, and support events.
Integration Monitoring	Tracks external service errors, retry state, processing backlogs, and operational failures.

5. Candidate Data Stores

Data Store	Architectural Purpose
Marketplace Order Store	Stores marketplace order records, buyer-accepted details, lifecycle state, totals, and references.
Seller Order Store	Stores seller fulfillment orders, seller status, assignments, and seller-level lifecycle state.
Order Item Store	Stores item-level status, price, quantity, seller, cancellation, shipment, and refund information.
Payment Store	Stores payment authorization, capture, void, refund status, gateway references, and payment state.
Tax Result Store	Stores tax calculations and supporting jurisdiction or tax breakdown details where available.

Data Store	Architectural Purpose
Availability Snapshot Store	Stores product and availability snapshots used during checkout and order acceptance.
Shipment Store	Stores carrier, tracking number, shipment status, shipment events, and delivery details.
Cancellation Store	Stores cancellation requests, eligibility results, reasons, actors, and outcomes.
Refund Store	Stores refund requests, calculated amounts, gateway responses, and final refund status.
Notification Store	Stores notification requests, delivery status, templates, and failures.
Support Case Store	Stores support notes, support actions, and optional case references.
Policy Configuration Store	Stores marketplace, seller, category, region, cancellation, refund, and support action rules.
Integration Status Store	Stores external request status, retry state, correlation IDs, failures, and reconciliation markers.
Audit Log Store	Stores significant user, system, and integration events.

6. Candidate External Interfaces

Interface	Provider	Consumer	Notes
Checkout Interface	Online Marketplace Order Platform	Buyer-facing marketplace experience	Supports checkout submission, validation, and order acceptance result.
Order Status Interface	Online Marketplace Order Platform	Buyer-facing marketplace experience	Supports buyer-visible order, shipment, cancellation, and refund status.
Seller Fulfillment Interface	Online Marketplace Order Platform	Seller portal or seller system	Provides seller orders and receives seller fulfillment updates.

Interface	Provider	Consumer	Notes
Support Order Interface	Online Marketplace Order Platform	Support experience	Supports order search, lifecycle review, support notes, cancellations, and refunds.
Payment Processing Interface	Payment Gateway	Online Marketplace Order Platform	Supports authorize, capture, void, and refund operations.
Tax Calculation Interface	Tax Calculation Service	Online Marketplace Order Platform	Supports tax calculation before checkout completion.
Shipping Rate and Tracking Interface	Shipping Carrier Service	Online Marketplace Order Platform	Supports rates, labels, tracking status, and shipment events where configured.
Notification Delivery Interface	Notification Provider	Online Marketplace Order Platform	Sends order lifecycle notifications.
Seller Inventory Interface	Seller System	Online Marketplace Order Platform	Confirms availability or reserves stock where supported.
Fraud Screening Interface	Fraud Screening Service	Online Marketplace Order Platform	Provides risk score, hold, approve, or reject result where configured.
User Authentication Interface	Identity Provider	Online Marketplace Order Platform	Authenticates seller, support, operations, and administrator users where configured.

7. Primary Workflows for Architecture Modeling

Workflow	Summary
Place Order	Buyer submits checkout; system validates cart, calculates total, authorizes payment, and creates order.
Validate Checkout	System verifies address, availability, tax, payment readiness, fraud status, and

Workflow	Summary
Split Seller Orders	marketplace policy. System decomposes a marketplace order into seller-specific fulfillment orders.
Confirm Seller Fulfillment	Seller accepts, rejects, or updates fulfillment responsibility.
Capture Payment	System captures authorized funds according to marketplace payment policy.
Track Shipment	System stores shipment details and updates delivery status from seller or carrier input.
Cancel Order	System validates eligibility, cancels affected order parts, and voids or refunds payment.
Issue Refund	Support or policy workflow submits full or partial refund to payment gateway.
Notify Participants	System sends buyer, seller, support, and operational notifications.
Resolve Integration Failure	System exposes failed external interactions and supports retry or manual intervention.
Review Order in Support	Support user searches order and reviews order, seller, payment, shipment, cancellation, and refund history.

8. Key State Models

Marketplace Order State

Candidate states:

1. Checkout pending.
2. Payment authorization pending.
3. Accepted.
4. Partially fulfilled.
5. Fulfilled.
6. Partially canceled.
7. Canceled.
8. Completed.
9. Support review.
10. Failed.

Seller Order State

Candidate states:

1. Created.
2. Sent to seller.
3. Awaiting seller response.
4. Accepted.
5. Rejected.
6. In fulfillment.
7. Partially shipped.
8. Shipped.
9. Delivered.
10. Exception.
11. Canceled.

Payment State

Candidate states:

1. Not authorized.
2. Authorization pending.
3. Authorized.
4. Authorization failed.
5. Capture pending.
6. Captured.
7. Voided.
8. Partially refunded.
9. Refunded.
10. Payment exception.

Item State

Candidate states:

1. Pending.
2. Confirmed.
3. Reserved.
4. Unavailable.
5. Fulfillment pending.
6. Shipped.
7. Delivered.
8. Canceled.
9. Refund pending.
10. Refunded.
11. Return pending.
12. Returned.

Integration Attempt State

Candidate states:

1. Pending.
2. Sent.
3. Succeeded.
4. Failed.
5. Retrying.
6. Timed out.
7. Requires manual review.
8. Abandoned.

9. Key Quality Drivers

Quality Driver	Architectural Implication
Consistency	Order, payment, seller, shipment, cancellation, and refund states must remain reconcilable.
Retry Safety	External operations must be safe to retry without duplicate orders, duplicate charges, duplicate refunds, or duplicate seller assignments.
Auditability	Lifecycle changes require durable actor, timestamp, reason, external reference, and prior-state records.
Reliability	External service failures must not leave the order in an ambiguous or unrecoverable state.
Observability	Operations users need visibility into failed payments, stuck seller orders, delayed shipments, and integration backlogs.
Security	Buyer, seller, support, and administrator access require different authorization boundaries.
Privacy	Buyer address, contact, payment references, seller details, and support notes require controlled access.
Scalability	Checkout, order creation, seller splitting, notifications, and shipment events must handle variable marketplace volume.
Performance	Checkout must remain responsive while

Quality Driver	Architectural Implication
Configurability	coordinating multiple external validations. Marketplace rules differ by seller, region, product category, fulfillment method, and support policy.

10. Architecture Risks

Risk	Architectural Concern
Payment-order inconsistency	Authorization, capture, void, and refund operations may succeed externally while local state update fails, or vice versa.
Multi-seller complexity	One buyer order may have multiple seller states, shipments, cancellations, and refunds.
Duplicate checkout submission	Buyer retries, browser refreshes, or network failures could duplicate order attempts unless controlled.
Seller non-response	Orders may become stuck if sellers fail to accept or reject fulfillment.
External tax dependency	Tax calculation failures can block checkout or produce inconsistent totals.
Partial cancellation and refund complexity	Item-level and seller-level changes must reconcile with payment and buyer-visible order state.
Carrier event variability	Shipment events may arrive late, duplicated, out of order, or not at all.
Fraud review timing	Fraud holds may conflict with seller fulfillment timing.
Policy sprawl	Seller-specific, region-specific, and category-specific rules may create hard-to-test state paths.
Support action authority	Support users may need controlled override powers without bypassing policy or audit requirements.

11. Open Architecture Questions

1. Is checkout allowed to create an order before payment authorization succeeds, or only after authorization?
2. When should payment capture occur: order acceptance, seller acceptance, shipment, delivery, or another policy point?

3. Are seller fulfillment orders created before or after payment authorization?
4. Is inventory reserved during checkout, after payment authorization, or after seller acceptance?
5. Which operations require strict idempotency keys?
6. How should the system reconcile payment gateway events that arrive after local timeout?
7. Should the platform support partial cancellation in the initial architecture?
8. Should refunds be item-level, seller-order-level, marketplace-order-level, or all three?
9. Are sellers required to use a portal, API, batch import, or email-based fallback?
10. Which external integration failures block checkout versus route to operational review?
11. How should shipment state be handled when sellers provide manual tracking numbers?
12. Are fraud screening holds required before seller notification?
13. How much marketplace policy should be configurable versus hard-coded in the first release?
14. What operational tooling is required to retry, reconcile, or manually resolve failed order workflows?

12. Recommended Modeling Scope

For the initial architecture model, focus on:

1. Checkout validation and order creation.
2. Payment authorization and payment state tracking.
3. Marketplace order to seller order splitting.
4. Seller fulfillment updates.
5. Buyer-visible order status.
6. Support order search and order detail view.
7. Cancellation before shipment.
8. Full refund handling.
9. Notification delivery.
10. Audit and integration status tracking.

Defer advanced modeling for:

1. Partial item cancellation.
2. Partial refunds.
3. Real-time carrier webhooks.
4. Fraud screening.
5. Seller system API integration.

6. Return merchandise authorization.
7. Seller payout reconciliation.
8. Advanced marketplace policy engine.